

POSTHARVEST MACHINERY STANDARDS & MOT's

1. Warranty of Mechanical Grain Dryer.

1 year

2. Max. Noise level of Mechanical Grain Dryer, Rice Mill, Corn Mill, Rice Reaper and Peanut Sheller

92 dB

3. Final Moisture Content required for Mechanical Grain Dryer

14% MCwb

4. Max. Moisture gradient for Mechanical Grain Dryer

2%

5. Static Pressure of Mechanical Grain Dryer

25 mmH₂O

6. Min. Drying Efficiency of Mechanical Grain Dryer

75%

7. Min. Heating System Efficiency for Petroleum based fuel Mechanical Grain Dryer

90% (direct-fired)

75% (indirect-fired)

8. Min. Heating System Efficiency for Biomass fuel Mechanical Grain Dryer

65% (direct-fired)

50% (indirect-fired)

9. For Mechanical Grain Dryer testing, the grain to be used shall be single variety and the moisture content shall be _____ and above for rice and corn with the highest available moisture content to be used in the test.

20%

10. For Batch dryers, at least _____ samples from the batch of grain to be dried shall be taken, spaced evenly over the loading period. Samples shall be taken from different portions of the dryer (bottom, middle and top).

12

11. For Mechanical Grain Dryer testing, particularly in the Standard laboratory method in assessing milling quality of paddy Palay sample is ____; Palay sample after cleaning is ____; and Brown Rice ____.

500 g ; 200 g ; 20 g

12 . It is a common type of thresher that gives off a capacity of 0.3 – 1.5 tons/hr
standard thresher

13a. Min. threshing efficiency for Mechanical Rice Thresher
99.8%

13b. Min. shelling efficiency for Corn Sheller
99.5%

13c. Min. shelling efficiency for Peanut Sheller
95%

14a. Min. threshing recovery for Mechanical Rice Thresher
97%

14b. Min. shelling recovery for Corn Sheller
97%

14b. Min. shelling recovery for Peanut Sheller
93%

15a. Max. Total losses for Mechanical Rice Thresher and Corn Sheller
3%

15b. Max. Total losses for Peanut Sheller
7%

16a. Max. Mechanically Damage grain for Mechanical Rice Thresher and Rice Combine
2%

16b. Max. Mechanically Damage kernel for Corn Sheller
3%

16c. Max. Mechanically Damage kernel for Peanut Sheller
3.5%

17. Max. Noise level for Mechanical Rice Thresher, Rice Combine and Fiber Decorticator.
95 dB

18. Based on **Mechanical Rice Thresher** Warranty from **PAES 204:2000**, **Rice Reaper** Warranty from **PAES 212:2004** and **Rice Combine Harvester** Warranty from **PAES 224:2005**, their construction shall be rigid and durable without major breakdown of the mechanisms within ____.
six (6) months

19. Preliminary operation of the machine to make various adjustments prior to the conduct of test

running-in period

20. Max. Grain Moisture Content for Mechanical Rice Thresher testing

24% MCwb

21. Straw Length for Mechanical Rice Thresher testing

45 cm – 50 cm

22. Grain Straw Ratio for Mechanical Rice Thresher testing

0.50-0.65

23. It is also known as *cargo rice*

brown rice

24. Type of a huller with rotating blades and utilizes pressure such as *Coriolis' force*.

centrifugal type

25. Min. Milling Recovery of Under Runner (*Cono*) Type Rice Mill

97%

26. Min. Milling Recovery of Rubber Roll & Centrifugal Type Rice Mill

98%

27. Min. Percent Head Rice of Rice Mill

90%

28. Min. Hulling Efficiency of Rice Mill

80%

29. Milling Degree of Rice Mill

Well-milled

30. No. of paddy per kilogram milled rice

15

31. Warranty of Rice Mill and Corn Mill.

6 months

32. The specified capacity at the brown rice output of the paddy separator must be attained, with maximum of ____pieces of palay per kilogram of brown rice output.

10

33. It is also known as *pearler*

Polisher

34. For rice mill, corn mill, and rubber roll testing, the sample's Grain Moisture Content must be ____.

14 ± 1%

35a. For rice mill, hammer mill, and rubber roll testing, the min. purity of sample is ____.

98%

35b. For corn mill testing, the min. purity of sample is ____.

95%

35c. Minimum Purity in Corn Sheller is ____.

98%

36. Palay whose average length of the full brown rice grain is above 6.5 millimeters
long grains

37. Palay with whole milled rice grain having length/width ratio over 3.0
slender grains

38. Type of shelling cylinder formed by a rolled metal sheet/plate or formed by longitudinal bars adjacently arranged forming a continuous cylinder.

closed-frame cylinder

39. Type of shelling cylinder where the shelling elements are attached to the equally spaced longitudinal bars arranged cylindrically.

open-frame cylinder

40. Type of shelling unit with cylinder having shelling elements of knife bar or Pegtooth
cylinder-type

41. Type of shelling unit consisting of a vertical disc with spiked surface

disc-type

42. Pistillate inflorescence of the plant *Zea mays L.*, enclosed with a leaf-like protective covering known as husk

corn ear

43. Unshelled fruit of the corn plant where the husk has been removed mechanically or manually

ear corn / dehusked corn / corn-in-cob

44. Max. Noise Level of Corn Sheller

100dB

45. For Corn Sheller testing, the sample's Moisture Content is ____.

18-24% MCwb

46. For Corn Sheller testing, the Kernel-Ear Corn Ratio of samples is ____.
0.77 – 0.81

48.a. The amount of test material to be supplied shall be sufficient for ____ of continuous **Threshing/Shelling Operation** including samples to be used for running-in prior to the actual conduct of test trials. A minimum of ____ test trials, with duration of at least ____ per trial, shall be adopted
one hour; two (2); 15 minutes

48.b. **PAES 202:2000 (Heated air mechanical grain dryer)** ____ trials shall be carried out with the same operational setting. The length of the test shall be such that one full capacity of grain has been dried to a final moisture content of approximately ____% (for rice and corn)
Three (3); 14%

49. Major component of the corn mill used to reduce corn kernels into corn grits.
roller mill

50. Min. Main Product recovery of Corn Mill
64%

51. Max. Total Losses of Corn Mill
5%

52. For Corn Mill Testing, the sample's variety to be used is ____.
Hybrid

53. Cutting mechanism consisting of fixed lower knife and reciprocating upper knife wherein its movement is controlled by the crank connected to the gear box or belt drive
reciprocating cutter knife

54. Cutting mechanism consisting of planetary type circular saw-toothed blade which rotates at the same time with the pick-up triangular frame
rotary knife

56. Grains that fall with the cut stalks during delivery and release at the side of the reaper during operation
conveying loss

57. Grains that have fallen to the ground due to the machine's cutting operation
header loss
shattering loss

58. Min. Field Efficiency of Rice Reaper.
65%

59. The rice reaper shall be operated using a minimum of _____ engine for a cutter bar cutting mechanism with _____ width of cut
3.5 hp (2.6 kW) and 1.2 m

60. A minimum of _____ engine shall be used for a rotary cutting knife cutting mechanism with _____ cutting width.
5 hp (3.7 kW) and 1.2 m

61. The reaper shall be operated at the speed of _____.
3 kph to 5 kph

62. Degree between the vertical line joining the center of the plant and the imaginary line where the stalk lodges
lodging angle

63. For rice reaper testing, each test, with three replications, shall be carried out in the rectangular field with an area of not less than _____ with minimum length of _____.
**750m²
30m**

64. The corners of the field should be manually harvested with the dimension of _____. to provide turning space for the machine
3 m x 3 m

65. For rice reaper testing, the suggested operational pattern is _____.
Circuitous pattern

66. It is the distance between two outermost divider tips.
Cutting width

67. The rubber for rubber rollers shall have a minimum thickness of _____.
15 mm

68. The rubber hardness shall be between _____ to _____ in the case of elastomers.
**80 shore
90 shore**

69. The rubber hardness shall be _____ for press-cured rubber mixes.
85 shore

70. When the rubber is subjected to heating test, the minimum value of hardness shall be _____ with maximum hardness drop of _____.
**70 shore
10 shore**

80. The tensile stress of the rubber roll shall be ____ to ____.

6 200 kPa | 8 268 kPa

81. The specific gravity of the rubber shall be ____ to ____.

1.2 | 1.3

82. The modulus of elasticity of the rubber shall be ____ to ____.

482 kPa

689 kPa

83. The rubber roll when tested in accordance with PAES 215 shall have hulling efficiency of at least ____.

80%

84. For every ____ grams reduction in weight of the rubber roll, the brown rice processed shall be at least ____ kg.

2 g

60 kg

85. This is carried out to obtain actual data on rubber roll performance, operating accuracy, work quality and adaptability to various paddy varieties for at least ____ hour/s of continuous operation or until there is a ____% reduction in the thickness of the rubber roll.

2 hr and 50%

86. One roll should rotate ____% faster than the other.

25%

87. Clearance adjustment between the two hullers should be maintained at about ____ the thickness of the paddy/rough rice.

$\frac{1}{2}$

88. It is the sum of the weight fractions retained above each sieve divided by 100.

Fineness Modulus

89. The hammer mill should be used for the desired product having a reduction ratio of at least ____.

50:1

90a. The noise emitted by the **hammer mill, forage chopper, and coconut decorticator** shall not be more than ____.

96 dB

90b. The noise emitted by the **large ruminant's slaughterhouse equipment**, particularly the **stunning box (knocking pen)** shall not be more than ____.

96 dB

90c. The noise emitted by **hog's slaughterhouse equipment**, particularly the **hog restrainer**, **dehairing machine** and **splitting saw for hog carcass** shall not be more than ____.

96 dB

91. For hammer mill testing, the locally grown corn must be used with max. MCwb of ____.

14%

92. For **hammer mill**, **forage chopper** and **peanut sheller** testing , the amount of test material to be supplied shall be at least ____ of input capacity of the machine.

75 %

93. This are devices intended to cut the crop into short lengths with reasonable consistency within a range of optional settings

Cutterhead / Cutting Rotor

94. It is a cylindrical roll generally with protrusions or flutes, used to gather, compress and advance the crop into the cutterhead.

Feedroll

95. This type of forage chopper where the chopped materials are thrown away to the front area of the machine

throw-away type

96. It is a type of forage chopper where the chopped materials are dropped down to the bottom of machine

let-fall type

97. It is a forage chopper which uses a feeding mechanism consisting of four or more feed rolls to partially orient and advance the crop at a consistent rate into the cutting mechanism

precision-cut forage chopper

98. This type of forage chopper is capable of producing the shortest and most uniformly cut particles.

precision-cut forage chopper

99. It is a forage chopper without a distinct feeding mechanism, usually employing flails to impact-cut and chop crop directly into shorter pieces.

random-cut forage chopper

100. This type of forage chopper usually produces the longest mean particle lengths, and the least uniformly cut particles.

random-cut forage chopper

101. It is a forage chopper which uses a feeding mechanism consisting of two feed rolls, or other means such as an auger, to advance the crop to the cutting mechanism

semi-precision-cut forage chopper

102. Mean particle lengths and particle uniformity are intermediate between those obtained with precision cut and random-cut forage choppers.

semi-precision-cut forage chopper

103. Precision-cut and semi-precision cut forage chopper shall be able to cut at least _____ settings with _____% maximum variation of cut.

Three different length

5%

104. Warranty against defective materials and workmanship shall be provided for parts except for normal wear and tear of consumable maintenance parts such as belts within _____ from the purchase of the **forage chopper / peanut sheller** and _____ warranty for services.

Six months

one year

105. Edible part of peanut

Kernel

106. Unbroken shell with kernel inside

Pod

107. Hull of the pod

Shell

108. For peanut sheller testing, the sample's MCwb must be _____.

12%-14%

109. Auxiliary conveyor to assist in feeding the crop to the threshing cylinder
crop elevator

110. Device which carries the grains from grain auger to grain tank or bin
grain elevator

111. Loss classified according to source, including all field losses attributable to the machine
grain loss

112. Pan for collecting the clean grains after being passed through cleaning sieve(s) for conveying to grain auger
grain pan

113. Tank used to hold the threshed grain
grain tank / bin

114. The _____ is a cleaning mechanism that oscillates at _____ in a plane that is parallel to the screen

oscillating screen

300 to 400 times per minute

115. Revolving slats or arms with battens arranged parallel to the cutter bar to hold the crop being cut by the knife and to push and guide it to a conveyor platform/feeder conveyor auger.

Reel

116. Portion of the combine comprising the mechanism for gathering, cutting, and picking the crop.

Header / Feed Table

117. Oscillating structure which supports the cleaning sieve(s) and which may also support the chaffer and the chaffer extension

shaker shoe / shoe

118. Assembly of two or more racks which agitates the straw and separates the remaining grains from straw

straw walker

119. Element placed on the rear side of the cylinder and above to rear ward of concave or Concave grate extension or transition grate to assist the deflection of straw on straw walker

stripper beater / rear beater

120. For combine harvester, the recommended reel speed should be _____ faster than the ground speed.

25 % to 50 %

121. For combine harvester, the recommended peripheral speed of the threshing drum _____.

20 m/s to 30 m/s.

122. Min. Field Efficiency of Rice Combine

75%

123. Total Rice Combine loss

3.5%

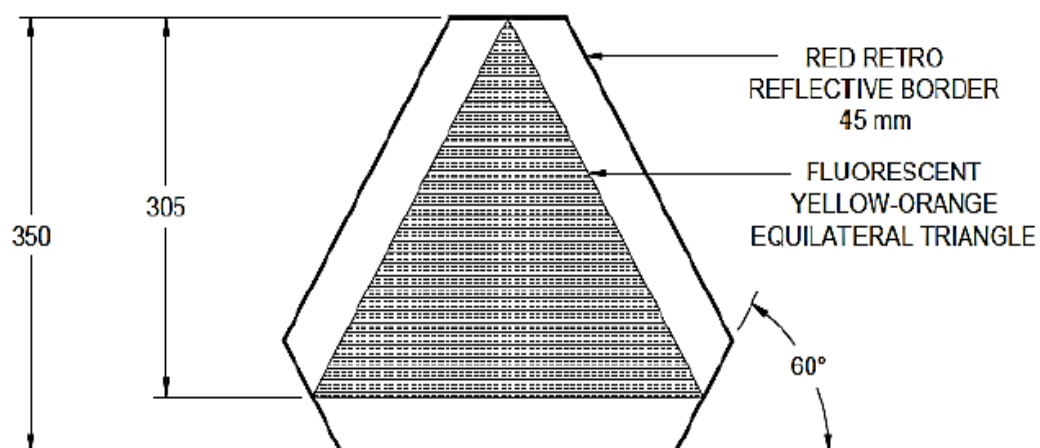
124. Slow-moving vehicle emblem has a _____ equilateral triangle inside and _____ reflective border at _____ mm.

Fluorescent Yellow-Orange; Red-Retro; 45

125. Slow-moving vehicle emblem has a total height of _____ and equilateral triangle height inside of _____.

350 mm

305 mm



Dimensions are in millimeter

Table 1. Performance Criteria for Mechanical Rice Thresher.

Criteria	Performance Data
Threshing Recovery, percent, minimum	97.0
Threshing Efficiency, percent, minimum	99.8
Losses, percent, maximum	
a) Blower Loss	1.2
b) Separation Loss	1.3
c) Unthreshed Loss	0.2
d) Scattering Loss	0.3
Purity, percent, minimum	
a) With Sifter and Fan	97.0
b) Without Sifter and With Fan	95.0
c) Without Cleaning Devices	80.0
Mechanically Damaged Grain, percent, maximum	2.0
Net Cracked Grain, percent, maximum	5.0
Noise Level, [db(A)], maximum	95.0*

* Allowable noise level for four (4) hours of continuous exposure based on Occupational Safety and Health Standards, Ministry of Labor, Philippines. 1983.